

DISCOVERY ISLAND, BC, Canada

WMO: 710310

Lat: 48.4247N

Lon: 123.2258W

Elev: 19

StdP: 101.10

Time Zone: -8.00 (INT)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.1	1.0	-11.1	1.5	0.5	-8.8	1.8	1.7	21.0	7.8	19.2	8.1	8.0	40	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.9	22.7	15.8	20.7	14.9	19.1	14.3	16.1	20.7	15.2	19.0	14.5	17.5	3.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.2	10.1	17.0	13.5	9.7	16.3	12.9	9.3	15.7	45.2	21.0	42.5	19.1	40.5	17.7	18.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	12.6	9.8	-2.9	27.3	2.9	2.4	-4.9	29.0	-6.6	30.4	-8.2	31.7	-10.3	33.4		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	6.7	6.8	7.9	9.7	12.0	13.9	15.2	15.5	14.3	11.3	8.6	6.6		
	DBStd	3.88	2.49	2.39	2.08	1.85	1.97	1.92	1.97	1.87	1.86	1.64	2.29	2.49		
	HDD10.0	456	104	91	70	26	3	0	0	0	0	6	51	106		
	HDD18.3	2789	361	323	324	258	196	136	102	93	123	219	292	362		
	CDD10.0	718	2	1	4	17	66	116	160	169	128	46	8	1		
	CDD18.3	11	0	0	0	0	0	2	4	3	1	0	0	0		
	CDH23.3	44	0	0	0	0	0	9	19	15	1	0	0	0		
	CDH26.7	5	0	0	0	0	0	1	3	1	0	0	0	0		
(14) Wind	WSAvg	4.1	5.5	5.2	4.4	3.6	3.2	3.2	3.4	3.2	3.0	3.8	5.1	5.9		
(15) Precipitation	PrecAvg	646	107	73	48	34	26	19	14	21	33	61	98	112		
	PrecMax	873	217	171	109	72	52	68	38	67	72	207	191	264		
	PrecMin	389	16	11	12	17	4	2	0	0	4	4	16	10		
	PrecStd	125	53	39	27	14	12	15	11	17	18	46	44	54		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	11.9	13.6	17.6	20.7	24.5	25.8	25.4	22.6	16.9	13.7	12.5		
		MCWB	10.7	9.5	9.5	11.0	12.4	15.5	16.0	15.9	16.1	13.8	11.5	10.0		
	2%	DB	11.1	10.6	12.0	14.7	18.1	21.1	22.1	22.4	20.2	15.3	12.5	11.2		
		MCWB	8.9	9.1	9.1	9.5	11.8	15.0	15.0	16.1	14.3	11.5	10.9	9.2		
	5%	DB	10.2	9.9	11.1	13.1	16.1	18.2	19.7	20.2	18.6	14.2	11.8	10.2		
		MCWB	8.6	8.3	8.7	8.9	11.6	13.7	14.8	15.4	13.5	11.1	10.5	8.5		
	10%	DB	9.5	9.3	10.3	12.0	14.7	16.6	18.1	18.6	17.3	13.3	11.2	9.5		
		MCWB	8.2	7.8	8.2	8.6	11.4	12.8	14.2	14.7	13.3	10.8	9.9	7.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	10.0	10.6	11.0	13.7	15.9	16.2	17.4	17.0	14.0	12.1	10.3		
		MCDB	10.1	10.9	12.1	13.6	18.5	20.1	20.1	23.2	21.9	15.3	12.9	11.5		
	2%	WB	8.7	9.1	9.4	9.9	12.6	14.2	15.1	16.5	15.4	12.3	11.2	8.9		
		MCDB	9.4	9.9	10.8	12.0	16.2	17.8	18.7	21.6	18.3	13.4	12.2	9.8		
	5%	WB	8.2	8.3	8.7	9.2	11.9	13.4	14.5	15.6	14.4	11.4	10.5	8.2		
		MCDB	9.1	9.1	10.3	10.9	14.7	16.3	17.7	20.2	16.5	12.6	11.4	9.1		
	10%	WB	7.5	7.5	8.2	8.7	11.3	12.7	13.9	14.7	13.6	10.8	9.8	7.5		
		MCDB	8.4	8.4	9.6	10.3	13.8	15.2	16.7	18.2	15.4	12.0	10.6	8.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	2.4	2.7	3.4	4.1	4.8	5.3	5.8	5.9	5.0	3.0	2.5	2.4		
		MCDBR	3.0	3.0	4.1	5.5	7.4	8.1	8.6	8.8	7.3	4.1	3.0	3.1		
	5% WB	MCWBR	2.5	2.1	2.5	3.0	3.6	4.1	4.0	4.6	3.7	2.6	2.6	2.6		
		MCWBR	2.9	2.7	3.9	4.2	6.2	6.7	6.9	8.7	6.2	3.8	3.1	2.8		
(40) Clear-Sky Solar Irradiance	taub		0.320	0.320	0.333	0.359	0.380	0.357	0.350	0.355	0.341	0.333	0.327	0.316		
		taud	2.521	2.543	2.521	2.439	2.398	2.481	2.536	2.520	2.559	2.562	2.501	2.483		
	Ebn at Noon		757	847	889	892	882	904	906	887	869	818	735	706		
		Edh at Noon	61	74	87	105	113	105	98	96	84	71	62	56		
(44) All-Sky Solar Radiation	RadAvg		0.93	1.81	2.83	4.29	5.48	5.99	6.55	5.52	3.94	2.17	1.11	0.75		
	RadStd		0.12	0.24	0.33	0.40	0.43	0.57	0.46	0.36	0.32	0.24	0.17	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		-0.80	-1.82	N/A	N/A	N/A	N/A	+151	+286	-149	N/A
(47) Regional (75 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page